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CPT: Differentiable Constraint Projection for Amortized, Zero-Shot Retargetable Steering of Autoregressive Generative Models

The per-step surrogate targets a concentrated distribution the sequence-level optimum does not

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ABSTRACT

Generative models are increasingly asked to produce outputs that satisfy a quantitative requirement rather than merely resemble their training data: a material at a target density, a molecule within a charge budget, a plan inside a cost. The standard formalization asks for the distribution closest to the model whose average meets the target, and that framing has a consequence practitioners rarely confront. An average says nothing about individual draws, so the cheapest route to it is to spread mass outward, and the resulting distribution can miss the requirement on most samples anyone would use. We take the other route, solving a small constrained projection of the next- token distribution at every decoding step and training through it. The projection is convex with an analytic derivative, making it a differentiable layer rather than a decoding heuristic. Two findings follow, on a task small enough to enumerate exactly and on crystal structure generation from a public materials database. First, the per-step route is not a worse approximation of the sequence-level solution but an approximation of a different distribution, and the difference favours it: at equal satisfaction of the average, the constrained quantity is 6.76 times less variable, and samples violate the requirement 1.42 times less often than samples from the distribution the field treats as optimal. Second, because the target enters as a value rather than a training condition, one trained network serves a family of targets, matching separately trained models at one seventh of their combined cost. Control of this kind is cheaper than its usual formalization suggests, and its limits are structural rather than incidental.

KEYWORDS controlled generation; constrained decoding; differentiable optimization layers; expectation constraints; crystal structure generation

1 Introduction

Materials discovery, molecular design and program synthesis increasingly depend on generative models, and in each the useful output is not merely a plausible one but one that meets a number: a lattice at a specified density, a composition inside a charge budget, a schedule within a cost. Learning the distribution of examples does not deliver this, because a model fit to a corpus reproduces the corpus's spread of the quantity rather than any value a user picks. Control of a generative model's quantitative behaviour is therefore a prerequisite for using one as a design tool at all.

The field's standard answer works at the level of whole sequences. Among all distributions whose expectation of a statistic meets the target, take the one closest in Kullback–Leibler divergence to the model; the solution is an exponential tilt of the base, and it is the fixed point that distributional policy gradient and its descendants pursue. Two properties of that answer are usually left implicit. It is defined over complete sequences, while an autoregressive model emits one token at a time, so reaching it requires either sampling from an unnormalized target or training against an estimator of it.

The second property is the one this paper turns on. An expectation constrains an average, which says nothing about any individual draw. Constrained optimization has distinguished these since chance-constrained programming separated expected from per-instance feasibility, and constrained reinforcement learning inherited the expectation form wholesale. For a generative model the distinction is not academic, because the samples a user inspects are individual draws rather than averages, yet the size of the gap between the two notions of feasibility is left unquantified in this setting.

Working per step is the obvious alternative and has been explored as decoding-time machinery: grammar and lexical constraints applied as masks, tractable surrogates consulted during search, and sequential Monte Carlo reweighting partial sequences. These act on a trained model's output rather than shaping what the model learns, so a model steered this way was never trained to expect the steering, and the mismatch between the distribution it learned and the one it is asked to emit is left for the sampler to absorb.

A separate literature supplies the missing ingredient. Convex programs with implicit derivatives can sit inside a network as layers, which is what a per-step constraint would need in order to be trained through rather than bolted on. That line and the decoding line have not been joined for expectation constraints, so how a per-step surrogate compares against the sequence-level optimum it approximates has never been measured against a model actually trained toward that optimum.

Our setting is crystal structure generation, where the target quantity is mass density and the serialization is text, alongside a synthetic task small enough that every distribution is computed exactly rather than sampled. Generative models for this domain are established enough that steering one is a fair test rather than a demonstration; Sec. 2 surveys them.

Contributions.

1. A differentiable per-step constraint projection with an analytic Jacobian, trained through rather than applied at decoding time (Sec. 3).
2. The concentration result: the surrogate targets a distribution whose constrained quantity is concentrated, while the sequence-level optimum anti-concentrates it (Sec. 4.2).

3. Zero-shot retargeting: one checkpoint covers a constraint family with statistically bounded equivalence to per-target training (Sec. 4.3).
4. Four scope boundaries, each measured rather than asserted: a steering ceiling, partial and asymmetric two-constraint composition, a serialization determinability limit, and inference-time baselines that beat the projection on accuracy at a diversity cost (Secs. 4.4 and 4.5).

> KEY TAKEAWAY
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> The per-step surrogate is not a cheaper approximation of the sequence-level optimum. It targets a different distribution, one whose constrained quantity is concentrated rather than spread, and on the tasks measured here that distribution is closer to what a request for a constraint usually means.

2 Related work

Sequence-level distributional control. Posing controlled generation as a moment constraint and solving for the KL-minimal distribution yields an exponential-family optimum, and distributional policy gradient trains a model toward it by importance-weighted sampling against that unnormalized target [1]. The formulation is attractive because the optimum is characterized in closed form up to a normalizer and the multiplier has a direct reading as the price of the constraint. Its practical difficulty is that the target is defined over complete sequences, so the training signal arrives only once a sequence terminates and the estimator's variance grows with the tilt strength. Where the resulting distribution is evaluated, it is evaluated on whether the constrained expectation is met, which is the quantity the formulation optimizes and not the quantity a user inspecting samples experiences.

Expectation versus per-instance constraints. The distinction our concentration result turns on is the one constrained Markov decision processes make between constraining an expectation and constraining every trajectory [2], refined by state-wise formulations [3] and by chance-constrained programming [4]. The precedent closest in mechanism is the safety layer, which projects a proposed action onto a feasible set at every step [5]. We adopt that vocabulary: the sequence-level tilt is an expectation constraint, and what our projection delivers is closer to a per-step feasibility guarantee whose aggregate happens to meet the expectation.

Per-step constrained decoding. Lexically constrained decoding enforces predicate-logic constraints during search without retraining [6], with lookahead heuristics added to reduce myopia [7]. GeLaTo instead distills a tractable generative model to supply exact conditional guidance [8], and sequential Monte Carlo methods reweight partial sequences against a target potential [9, 10]. All three act at the per-step locus but treat the constraint as a filter over candidates or a reweighting of them, so none yields a distribution that is itself the solution of a constrained problem, and none is differentiable with respect to the constraint. Ahmed et al. [11] are the nearest training-time neighbour, relaxing a logical constraint into a pseudo-semantic loss over a local approximation, the same shape of move as reducing a sequence-level expectation to a per-step one, though approximate in the constraint rather than exact in the support.

Differentiable optimization layers. Embedding an argmin in a network and backpropagating through its solution is established for quadratic programs, where differentiating the Karush–Kuhn–Tucker conditions gives the gradient in closed form [12]; for general convex programs, where the same idea is automated by rewriting the problem in a canonical cone form [13]; and for hard constraints, where a completion-plus- correction scheme restores feasibility that a learned map would otherwise violate [14]. What these share is a cost profile: a generic solver is called in the forward pass and its implicit derivative in the backward pass, so the layer is affordable per example but not per token. None of them has been instantiated as a divergence projection over a next-token distribution, where the problem is small and structured enough that a generic solver is the wrong tool.

Crystal structure generation. Generative models for inorganic crystals include diffusion over invariant representations [15, 16], autoregressive language models over text serializations [17, 18], and large-scale diffusion trained for property targeting [19]. We use the same MP-20 subset of the Materials Project [20] and a text serialization in the spirit of the autoregressive line, because a serialization is what makes a per-step budget definable at all. That the class is bounded this way is a real constraint on the method rather than an implementation detail.

3 Method

Let $p_\theta(\cdot \mid x_{<t})$ be the model's next-token distribution over a vocabulary $\mathcal{V}$, let $m_t \in \{0,1\}^{|\mathcal{V}|}$ be a mask marking tokens the grammar permits after the prefix $x_{<t}$, and let $\phi \in \mathbb{R}^{|\mathcal{V}|}$ assign each token an increment of the constrained quantity. Given a target total c for a sequence of length L, define the running budget

$$b_t = \frac{c - \sum_{s<t} \phi_{x_s}}{L - t + 1}, \tag{1}$$

the per-step increment that would land the total on target if every remaining step matched it. The layer solves

$$q_t = \arg\min_{q \in \Delta(\mathcal{V})} \mathrm{KL}(q \,\|\, p_\theta(\cdot \mid x_{<t})) \quad \text{subject to} \quad \langle q, \phi \rangle = b_t, \quad \mathrm{supp}(q) \subseteq \mathrm{supp}(m_t). \tag{2}$$

This is a KL projection onto the intersection of an affine slice with a face of the simplex. Its solution is an exponential tilt of the masked model distribution, $q_t \propto m_t \odot p_\theta \exp(\lambda\phi)$, with the multiplier λ fixed by the constraint; a one-dimensional monotone root find gives it to machine precision. Sec. S1 verifies the solve and its gradient oracle. Two properties matter downstream. The support restriction is satisfied *exactly*, because masked tokens receive zero mass by construction. The expectation constraint is satisfied exactly *when it is attainable*: if b_t lies outside the range of ϕ over the allowed tokens, no distribution on that support meets it, and the layer falls back to the unconstrained masked distribution and records the fallback.

Differentiating the solve. The Jacobian of q_t with respect to the incoming logits follows from the implicit function theorem applied to the scalar stationarity condition, and reduces to a rank-one correction of a diagonal, so the backward pass costs a few vector operations rather than a linear solve. Because the correction involves only inner products with ϕ, the Hessian-vector products

needed for the multiplier's sensitivity can be batched; we measure a speedup from single precision of 1.33× to 2.64× over the batch sizes tested (Sec. S2).

What the mask does and does not do. On the crystal task the mask is a character-level grammar over a text serialization. It removes the syntactic failure mode completely, since every sampled string parses, but it is not a validity oracle; Sec. S3 separates the two effects.

4 Experiments and results

4.1 Shared setup

Two settings carry every result below, and both are described here once rather than repeated per experiment. The *synthetic* setting is a length-8 sequence over a 4-symbol vocabulary, so all 65,536 sequences enumerate and every distribution is computed rather than sampled: no divergence reported on this task is an estimate. A per-symbol multiplicity budget defines the support constraint, satisfied by 36,120 sequences, and each symbol carries a charge; the expectation constraint asks for a mean total charge of zero. The *crystal* setting is the MP-20 subset of the Materials Project, serialized as text under the grammar of Sec. 3, where the constrained quantity is mass density in g/cm^3.

Four training arms recur throughout, differing only in what they train through rather than in architecture or mask. The *mask-only* arm trains with the grammar mask and no projection. The *projection* arm trains through Eq. (2) and is read either without the layer, giving its backbone, or with it, giving the deployed system. The *sequence-level comparator* trains by maximum likelihood on exact samples from the sequence-level tilt, which is distributional policy gradient's fixed point with unit importance weights; that rule is chosen deliberately, because it gives the comparator the best case of the method it stands for, so no gradient-estimator artefact can be blamed for the outcome. Baselines needing no training are named where they are used.

Metrics follow the constraint. On the synthetic task we report the mean of the constrained quantity, its variance, and the per-sequence violation rate, meaning the probability mass on sequences whose own total misses the target. On the crystal task we report absolute density error, structural validity under an independent parser, and unique-formula fraction as a diversity measure. Every figure reported below is read from a stored artifact at build time rather than transcribed.

4.2 The surrogate targets a concentrated distribution

The comparator reaches its target. The comparator lands 0.0167 ± 0.0048 nats from the exact tilt with a mean charge of -0.062 ± 0.122 against a target of zero and no mass outside the mask. It is not a failed run, which is what makes the comparison in Table 1 informative rather than a win over something broken.

Result. Both the projection and the comparator satisfy the expectation constraint, and the comparator's seed spread on mean charge is wider than the gap between them, so we claim no ordering there. The difference that lies outside seed noise is in the last two columns. The optimum's charge variance exceeds the mask-conditioned data's, *raising* the variance of the quantity being constrained by 22.1%, while the projection lowers it by 81.9%. Against the optimum rather than against the unconstrained arm, the projection's variance is lower by a factor of 6.76, a gap exceeding three of

Table 1 | Both arms satisfy the expectation constraint; only one concentrates it. The sequence-level optimum's charge variance exceeds the unconstrained data's, while the projection's is far below it. Per-sequence violation is the mass on sequences whose own total charge is nonzero. Mean ± s.d. over seeds where an arm is trained; the tilt and the data distribution are exact.

Arm	Mean charge	Violation	Variance
Data, mask-conditioned (exact)	2.5164	0.9109	8.564
Mask only	2.4694	0.9083	8.762
Sequence-level comparator	−0.0619	0.8751	10.362
Tilt, sequence-level optimum	0.0000	0.8755	10.452
Projection (deployed)	**+0.0170**	**0.6155**	**1.547**

the comparator's seed standard deviations. Individual samples follow: the optimum's own mass sits on constraint-violating sequences 1.42 times more often than the projection's does.

The mechanism is visible in Fig. 1 and follows from what the tilt is. It is the KL-closest distribution with the right mean, and reaching a mean of zero from a base centred at +2.469 is cheapest by moving mass symmetrically outward. Nothing in the objective rewards concentration. Bounding an expected increment at every step, by contrast, bounds the total in expectation and narrows it as a side effect, because each step's feasible set is centred on what the remaining budget can still absorb. The surrogate's 1.025 nats of excess divergence from the optimum is therefore a trade rather than a loss, and Sec. S4 reports how that overhead varies with constraint strength.

4.3 One checkpoint retargets across the constraint family

Because the constraint enters Eq. (1) as a target value rather than as a training condition, a trained checkpoint can be deployed at values it never trained for. We test this on the crystal task, where the constrained quantity is mass density and the family is ten density targets spanning the data distribution. Three checkpoints, each trained at one target, are deployed across all ten. Energies quoted here come from a surrogate (Sec. S5), not density functional theory.

Structural validity is exact at every one of the thirty cells, including targets no checkpoint trained for and targets beyond the reachable range. The mask is a property of the grammar, not of the training target, so retargeting cannot break it.

On accuracy, one checkpoint deployed across the sub-ceiling family is level with the frontier of 7 separately trained models. A difference test is the wrong instrument here. A large p-value is absence of evidence, not evidence of equivalence, and at $n = 7$ paired targets the smallest attainable two-sided Wilcoxon p is $2/2^7 = 0.0156$ under any data, which Sec. S6 verifies by exhaustive enumeration. We therefore pre-registered an equivalence margin, defined as the retargeted arm's own mean within-target seed standard deviation, on the reasoning that a difference below the method's own reproducibility is not actionable and that a margin tied to the arm's own noise cannot be widened to pass without changing the arm. Two one-sided tests against that margin of 0.189 give $p = 0.0023$, and the pre-registered secondary condition ($p < 0.05$ at a 0.10 margin) also holds (Sec. S7).

The honest form of the claim is parity, not superiority: on a single seed the retargeted checkpoint

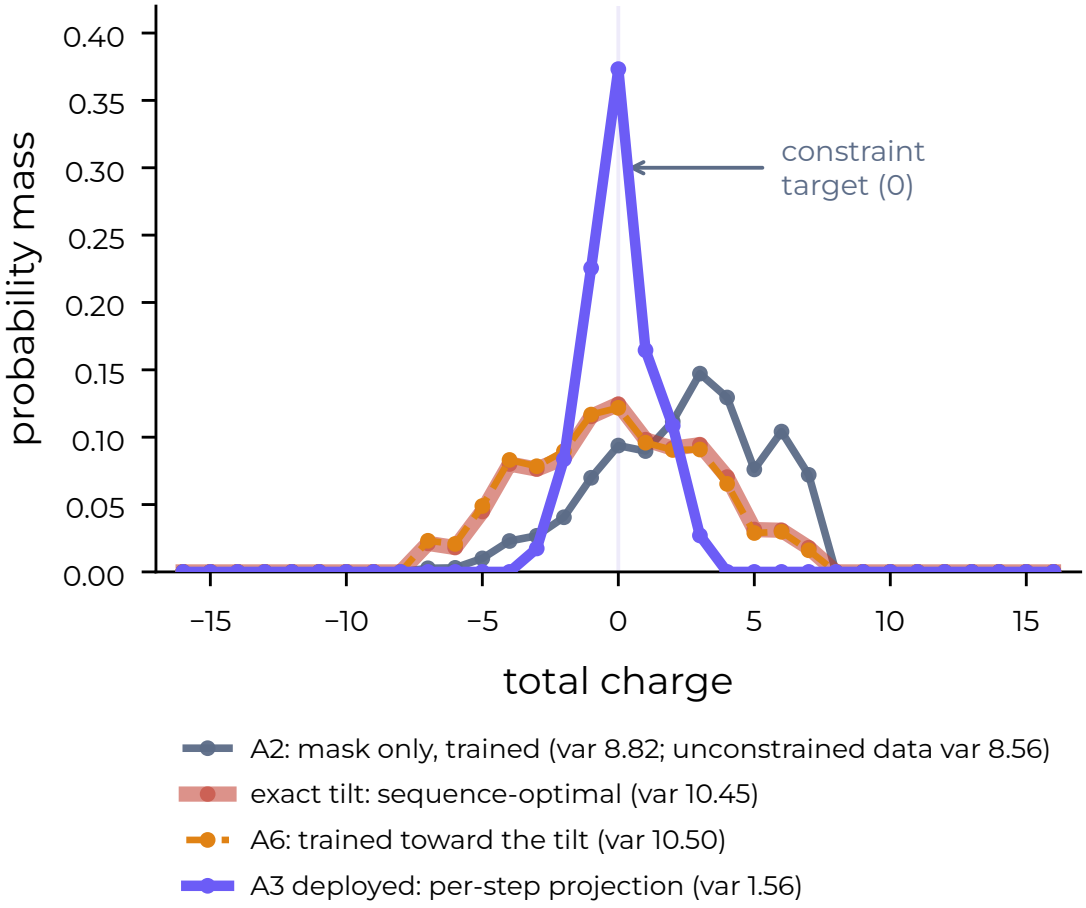


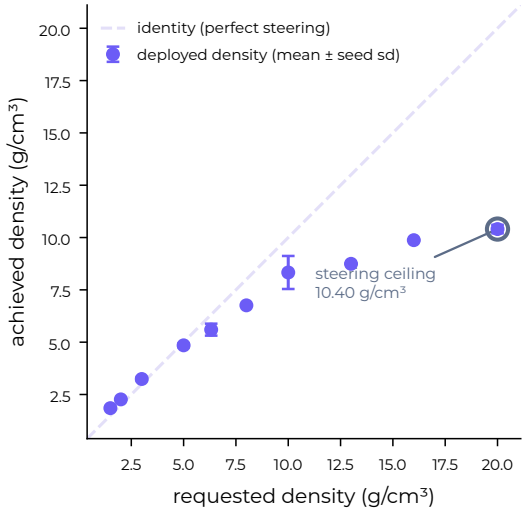
Figure 1 | The optimum spreads the constrained quantity; the projection concentrates it. Exact charge distributions over all sequences. The sequence-level optimum and the model trained toward it place mass at both extremes to buy a mean of zero, ending wider than the mask-only arm they started from. The projection instead spikes at the target. Legend entries give each arm's charge variance.

appeared *better* than the frontier, but replication across seeds found it better at only 3 of 7 targets. What survives is that one training run is statistically indistinguishable from seven at one seventh of the training compute, which is a claim about cost structure rather than about accuracy. Table S1 gives the per-seed values behind both readings.

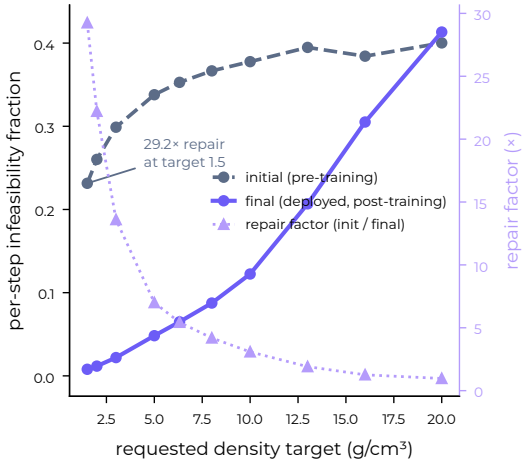
4.4 Where the method stops

Steering saturates at a backbone ceiling. Requested and achieved density track closely at low targets and then flatten (Fig. 2a). The plateau sits at 10.405 g/cm^3: requests beyond it are not met, and no amount of additional tilt reaches them, because the projection can only redistribute mass the backbone already assigns. The constraint is a reweighting of the model's own proposals, not a source of new modes.

Fig. 2b gives the mechanism. Before training, a large fraction of decoding steps present a budget



(a). Requested against achieved density, with the ceiling marked.



(b). Infeasible steps before and after training, with the repair factor.

Figure 2 | The ceiling and its mechanism. Steering follows the request until the backbone runs out of mass to redistribute; the repair the layer performs weakens as the request hardens, and where it approaches unity the ceiling binds.

outside the attainable range, and the layer falls back. Training repairs most of that at easy targets and progressively less at hard ones; where the repair factor approaches unity, the ceiling binds.

Two constraints compose partially, and asymmetrically. The second additive constraint is valence electron density, chosen because it is computable for every element in the split and only partly correlated with mass density ($r = 0.725$). The joint solve converges but steers weakly. Across 7 trained cells, the fraction of the requested valence shift that survives to the output never exceeds 11.3%, and the second constraint's effect is dominated by budget exhaustion: at most positions the mass budget the first constraint has already committed leaves nothing for the second to allocate.

The asymmetry is sharper than the aggregate. Asked to *lower* valence density the constraint moves the right way in 2 of 2 cells; asked to *raise* it, in 2 of 5. A running budget is a remaining quantity: spending less of it is always available by choosing lighter or fewer atoms, while spending more than the prefix has left is not available once the budget is exhausted. Sec. S8 gives every cell against its own control. Charge neutrality, the constraint one would most want here, is not expressible in this class at all: it quantifies existentially over oxidation states, and collapsing to one charge per element destroys the predicate rather than approximating it (Sec. S9).

The constraint class depends on the serialization. A per-step budget requires that the prefix determine it. Mass density is a ratio, so a target density fixes a mass budget only once the cell volume is known. Over 393 structures and 1,217 element positions, a serialization that closes the cell parameters first makes the budget determinable at 100% of positions, while one that emits composition first makes it determinable at 0%. The constraint class is therefore a property of the serialization, not of the layer, and a target quantity that no prefix determines cannot be steered by

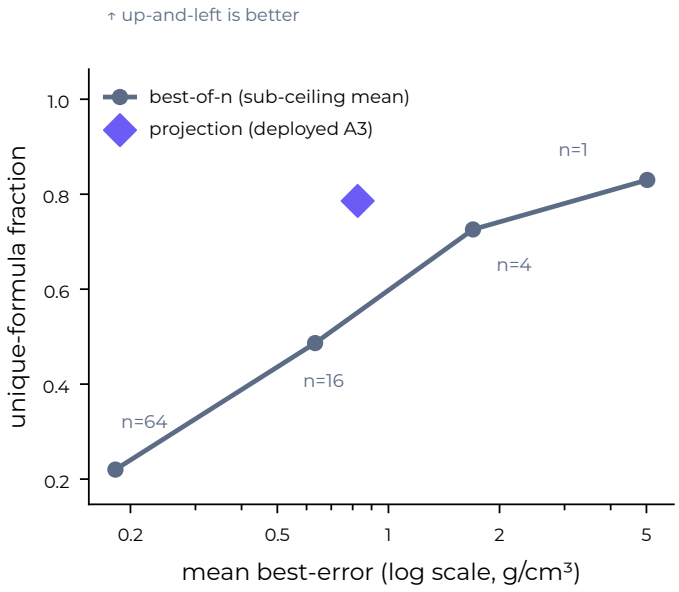


Figure 3 | **Accuracy bought by selection is diversity spent.** Mean best-error against unique-formula fraction over the sub-ceiling targets. Best-of-n walks down and to the left as n grows, trading output variety for accuracy; the projection sits off the curve at one forward pass per output.

this mechanism at all.

4.5 Baselines

Conditional training is more accurate, where it has data. Training a single model conditioned on the target value is the strongest same-cost baseline. It is more accurate than the projection wherever the training split has support: mean absolute error 0.085 against 0.196 below the data mean. It pays for that in coverage, emitting 5.9% novel formulas against 94.1%, and its accuracy degrades where the conditioning slice thins, precisely where a practitioner asking for an unusual target is operating.

Best-of-n beats the projection on accuracy, and pays in diversity. Given enough inference compute, drawing n samples from the unsteered model and keeping the closest one needs no training and wins on accuracy: it crosses the projection's error at $n = 16$ (Fig. 3). The cost is concentration of the accepted set. Unique formula fraction falls from 0.824 at $n = 1$ to 0.228 at $n = 64$, while the projection's 0.782 does not depend on how hard the target is to hit. Sequential Monte Carlo, the principled member of the same family, reaches the projection at 6 of 10 targets within $K = 64$ particles but is worse than best-of-n at equal compute by a mean factor of 4.2; its potential width is untuned, so that figure is a lower bound on what SMC could achieve rather than a refutation of it.

The honest reading is that the projection's advantage is not accuracy per target. It is that the cost is paid once at training and never again at inference, while the inference-time family pays per output forever and concentrates its outputs to do so.

5 Discussion

The result we did not expect is that the sequence-level optimum anti-concentrates the quantity it constrains. Stated after the fact it is almost obvious, since an expectation constraint says nothing about individual draws and the cheapest way to move a mean is to move the tails. It nonetheless inverts the usual reading of a per-step surrogate as a compromise. The distinction between constraining an expectation and constraining every trajectory is old; what we add is a measurement of how far apart those two objects are for a concrete generative model, and the observation that the per-step route lands closer to the second.

The boundaries are as informative as the results. The steering ceiling says the projection reweights rather than creates. The composition asymmetry says a running budget is a remaining quantity, so constraints that want more of it fail differently from constraints that want less. The determinability result says the expressible constraint class is chosen when the serialization is chosen, which is a design decision usually made for other reasons. And best-of-n's accuracy win says the contribution is cost structure, not accuracy: a claim weaker than the one we set out to make and better supported than it would have been. Sec. S10 records three errors caught during this work and the mechanism that caught each, since each would otherwise have produced a plausible-looking wrong number.

6 Conclusion

Solving a small differentiable constrained projection at every decoding step, and training through it, produces a model whose samples individually respect a quantitative target far more often than samples from the distribution the field treats as optimal for that target. That inversion is the paper's substantive finding, and it has a mechanism rather than a coincidence behind it: an expectation constraint is indifferent to individual draws, so the cheapest route to a mean is through the tails, while a per-step feasible set is centred on what the remaining budget can absorb. Because the target enters as a value rather than a training condition, the same trained network serves an entire family of targets, which relocates the cost of control from inference to a single training run.

Limitations. The projection reweights the backbone's own proposals, so it cannot reach a target the backbone assigns no mass to, and the achievable range saturates well before the requested one does. Two correlated constraints sharing a single token budget compose only partially, and asymmetrically: the direction that spends less of the budget is reliably available while the direction that spends more is not. The expressible class is fixed by the serialization, since a target no prefix determines cannot be steered at all, and charge neutrality falls outside that class rather than merely being unimplemented. On accuracy per target the method is not the strongest option available, because sampling repeatedly from an unsteered model and selecting the closest output wins given enough inference compute. The synthetic results are exact but small, and the crystal energetics rest on a surrogate rather than density functional theory.

Future work. Three threads follow directly. The two-constraint grid is one row short of the range needed to separate a smooth dependence on target compatibility from a jump, and completing it is a matter of compute rather than method. The particle-filter comparison used an untuned potential width, so the figure reported here bounds that family from below rather than characterising it,

and a tuned version would sharpen the comparison the paper draws. The serialization result is currently a statement about two orderings measured on one quantity; turning it into a predictive account of which constraints a given serialization admits needs several serializations budgeted together, which is the form in which it would become a design rule rather than an observation.

> **REPRODUCIBILITY**
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> Every number here is read from a JSON artifact in `results/` at build time, never transcribed, and each figure and load-bearing value carries a comment naming its artifact; the test suite recomputes them and fails if the prose disagrees.

AUTHOR CONTRIBUTIONS

C.P. designed the layer, ran the experiments, and drafted the paper. H.K. supervised the project and revised the manuscript.

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DATA AND CODE AVAILABILITY

Code and artifacts: https://github.com/KurbanIntelligenceLab/cpt. The MP-20 subset used here is public.

COMPETING INTERESTS

The authors declare no competing interests.

References

[1] Muhammad Khalifa, Hady Elsahar, and Marc Dymetman. A distributional approach to controlled text generation. In *International Conference on Learning Representations (ICLR)*, 2021.

[2] Eitan Altman. *Constrained Markov Decision Processes*. 1999.

[3] Weiye Zhao, Rui Chen, Yifan Sun, Ruixuan Liu, Tianhao Wei, and Changliu Liu. State-wise constrained policy optimization, 2023. Preprint; withdrawn ICLR 2024 submission.

[4] A. Charnes and W. W. Cooper. Chance-constrained programming. *Management Science*, (1), 1959. doi: 10.1287/mnsc.6.1.73.

[5] Tu-Hoa Pham, Giovanni De Magistris, and Ryuki Tachibana. OptLayer – practical constrained optimization for deep reinforcement learning in the real world. In *2018 IEEE International Conference on Robotics and Automation (ICRA)*, 2018. doi: 10.1109/ICRA.2018.8460547.

[6] Ximing Lu, Peter West, Rowan Zellers, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. NeuroLogic decoding: (un)supervised neural text generation with predicate logic constraints. In *Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, Online, 2021. doi: 10.18653/v1/2021.naacl-main.339.

[7] Ximing Lu, Sean Welleck, Peter West, Liwei Jiang, Jungo Kasai, Daniel Khashabi, Ronan Le Bras, Lianhui Qin, Youngjae Yu, Rowan Zellers, Noah A. Smith, and Yejin Choi. NeuroLogic a*esque decoding: Constrained text generation with lookahead heuristics. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*, Seattle, United States, 2022. doi: 10.18653/v1/2022.naacl-main.57.

[8] Honghua Zhang, Meihua Dang, Nanyun Peng, and Guy Van den Broeck. Tractable control for autoregressive language generation. In *International Conference on Machine Learning (ICML)*, 2023.

[9] Stephen Zhao, Rob Brekelmans, Alireza Makhzani, and Roger Baker Grosse. Probabilistic inference in language models via twisted sequential Monte Carlo. In *International Conference on Machine Learning (ICML)*, 2024.

[10] João Loula, Alexander K. Lew, Tan Zhi-Xuan, Vikash Mansinghka, Tim Vieira, Timothy J. O'Donnell, Ryan Cotterell, and Jason Eisner. Syntactic and semantic control of large language models via sequential Monte Carlo. In *International Conference on Learning Representations (ICLR)*, 2025.

[11] Kareem Ahmed, Kai-Wei Chang, and Guy Van den Broeck. A pseudo-semantic loss for autoregressive models with logical constraints. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023.

[12] Brandon Amos and J. Zico Kolter. OptNet: Differentiable optimization as a layer in neural networks. In *International Conference on Machine Learning (ICML)*, 2017.

[13] Akshay Agrawal, Brandon Amos, Shane Barratt, Stephen Boyd, Steven Diamond, and J. Zico Kolter. Differentiable convex optimization layers. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[14] Priya L. Donti, David Rolnick, and J. Zico Kolter. DC3: A learning method for optimization with hard constraints. In *International Conference on Learning Representations (ICLR)*, 2021.

[15] Tian Xie, Xiang Fu, Octavian-Eugen Ganea, Regina Barzilay, and Tommi S. Jaakkola. Crystal diffusion variational autoencoder for periodic material generation. In *International Conference on Learning Representations (ICLR)*, 2022.

[16] Rui Jiao, Wenbing Huang, Peijia Lin, Jiaqi Han, Pin Chen, Yutong Lu, and Yang Liu. Crystal structure prediction by joint equivariant diffusion. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[17] Luis M. Antunes, Keith T. Butler, and Ricardo Grau-Crespo. Crystal structure generation with autoregressive large language modeling. *Nature Communications*, 2024. doi: 10.1038/s41467-024-54639-7.

[18] Nate Gruver, Anuroop Sriram, Andrea Madotto, Andrew Gordon Wilson, C. Lawrence Zitnick, and Zachary Ulissi. Fine-tuned language models generate stable inorganic materials as text. In *International Conference on Learning Representations (ICLR)*, 2024.

[19] Claudio Zeni, Robert Pinsler, Daniel Zügner, Andrew Fowler, Matthew Horton, Xiang Fu, Zilong Wang, Aliaksandra Shysheya, Jonathan Crabbé, Shoko Ueda, Roberto Sordillo, Lixin Sun, Jake Smith, Bichlien Nguyen, Hannes Schulz, Sarah Lewis, Chin-Wei Huang, Ziheng Lu, Yichi Zhou, Han Yang, Hongxia Hao, Jielan Li, Chunlei Yang, Wenjie Li, Ryota Tomioka, and Tian Xie. A generative model for inorganic materials design. *Nature*, 2025. doi: 10.1038/s41586-025-08628-5.

[20] Anubhav Jain, Shyue Ping Ong, Geoffroy Hautier, Wei Chen, William Davidson Richards, Stephen Dacek, Shreyas Cholia, Dan Gunter, David Skinner, Gerbrand Ceder, and Kristin A. Persson. Commentary: The Materials Project: A materials genome approach to accelerating materials innovation. *APL Materials*, (1), 2013. doi: 10.1063/1.4812323.

[21] Bowen Deng, Peichen Zhong, KyuJung Jun, Janosh Riebesell, Kevin Han, Christopher J. Bartel, and Gerbrand Ceder. CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling. *Nature Machine Intelligence*, 2023. doi: 10.1038/s42256-023-00716-3.

[22] Bowen Deng, Janosh Riebesell, Kevin Han, Luis Barroso-Luque, and Peichen Zhong. CHGNet v0.3.0, 2023. Software release, v0.3.0, used as the energy-screening surrogate in this work.

APPENDIX

S1 Solver verification and the gradient oracle

The projection defined in the method is solved by a one-dimensional monotone root find on the multiplier. Two things are verified rather than assumed. First, the solve converges to the constraint to machine precision wherever the target is attainable, and reports non-convergence rather than returning a near-miss where it is not; the non-convergence flag is what the fallback path keys on. Second, the analytic Jacobian agrees with finite differences and with autograd through an unrolled solver.

For the multi-constraint form, convergence alone is insufficient: a converged solve on a near-singular system is not trustworthy. We therefore report the conditioning of the constraint matrix alongside every joint solve, and sweep vocabulary size rather than testing a single configuration, because a single grid at one vocabulary would have read as "constraints always compose".

S2 Overhead and timing

Training through the projection costs 6.0× the wall clock of training with the mask alone, on the synthetic task at matched steps. The backward pass is a rank-one correction rather than a linear solve, and batching the Hessian-vector products in single precision gives a speedup ranging from 1.33× at the smallest batch to 2.64× at the largest. At inference the layer adds one root find per step.

S3 Grammar mask and parse validity

The character-level grammar makes every sampled string parse: the syntactic parse rate is 1.00 under an independent parser. It is not a validity oracle: only 47% of those strings build into a structure, the remainder failing on semantic grounds such as overlapping sites or unphysical stoichiometries, which parse without complaint. The mask leaves real freedom: a mean of 4.14 allowed characters per step, though 70.6% of steps are fully forced by the grammar.

S4 How the overhead varies with constraint strength

The surrogate's excess divergence over the sequence-level optimum is close to fixed in absolute terms while the optimum's own cost varies over a 129-fold range as the constraint pulls harder. The relative waste therefore falls from 97.8% at the weakest steering to 32.4% at the strongest: the surrogate is least efficient exactly where the constraint barely matters, and most efficient where it binds. Sec. S1 covers the solve itself and Sec. S2 its wall-clock cost.

S5 Energy screen calibration

Formation energies are estimated with CHGNet [21] at version v0.3.0 [22] rather than density functional theory. On 1,456 held-out structures the surrogate agrees with its reference at 0.0210 eV/atom mean absolute error and Pearson $r = 0.9996$. This is a surrogate screen and is labelled as such wherever it is used; no claim in the main text depends on a DFT-quality energy.

Table S1 | Per-seed retargeting summary. Densities in g/cm^3. Seed 0's backbone is an excursion, which is why its lower error is not the reading to take; the published ceiling is 10.405. Structural validity is 1.0 in every cell of every seed.

Seed	Backbone mean	Max achieved	Sub-ceiling MAE
0	22.769	15.495	0.4406
1	8.381	10.439	0.7889
2	8.692	10.596	0.7197
Mean of seeds	–	–	0.6497
Frontier, 7 runs	–	–	0.6620

S6 Statistical floor of the difference test

The claim that a paired Wilcoxon test at $n = 7$ cannot return $p < 0.0156$ is verified by exhaustive enumeration over all 2^7 sign patterns rather than by appeal to a table. This is why the equivalence procedure, rather than the difference test, carries the retargeting claim.

S7 The full equivalence appendix

The margin is defined before the test, in a pre-registration file that the analysis script reads; the script fails if the file is absent or if its value disagrees with the stated rule. The rule is that the margin equals the retargeted arm's mean within-target seed standard deviation. Two one-sided tests give $p = 0.0023$ at that margin of 0.189, and the secondary pre-registered condition, still $p < 0.05$ at a 0.10 margin, also holds. A bootstrap interval is reported alongside the parametric one and the wider of the two is what the main text quotes.

The difference test is reported for completeness and interpreted as uninformative: the paired Wilcoxon over 7 targets returns $p = 1.0$, but the smallest attainable two-sided p at that sample size is $2/2^7 = 0.0156$ under any data whatsoever, so a large value carries almost no information.

The table below gives the three replication seeds behind both readings, and it is included because the aggregate hides the reason to distrust one of them. Each row is a separate training run of the projection arm at the same target, read first without the layer to give its backbone and then with the layer across the whole sub-ceiling family. The backbone column is what makes seed 0 unusable as evidence: at 22.769 it sits far above the other two, which land at 8.381 and 8.692, so its lower sub-ceiling error of 0.4406 reflects a backbone that happened to propose high-density structures rather than a layer that steered better. Averaging the three gives 0.6497 against the frontier's 0.6620, which is the comparison the main text reports, and the spread of 0.1506 across seeds is what the pre-registered margin is derived from. All three seeds exceed the published ceiling of 10.405, so the excursion is a matter of degree rather than of kind.

S8 Two-constraint grid, per cell

Table S2 gives every cell of the two-constraint grid at a shared valence target, each against its own single-constraint control. The control is necessary: the achieved valence density rises with the density target even when no valence constraint is applied, so an absolute level would show

Table S2 | Direction of the second constraint's effect. Cells at a shared valence target across density rows. "Asked" is the direction the target sits relative to that cell's control. Lowering the quantity succeeds in both cells that request it; raising it succeeds in two of five.

Valence	Density	Achieved	Control	Deviation	Asked	Correct
0.282	3.0	0.2447	0.2424	0.0023	up	yes
0.282	5.0	0.3062	0.3141	−0.0079	down	yes
0.282	8.0	0.3251	0.3956	−0.0705	down	yes
0.484	3.0	0.2603	0.2424	0.0179	up	yes
0.484	8.0	0.3482	0.3956	−0.0474	up	no
0.686	3.0	0.2353	0.2424	−0.0071	up	no
0.686	8.0	0.3707	0.3956	−0.0248	up	no

apparent steering where there is none. Only the deviation from control is the second constraint's effect.

S9 The boundary of the expressible class

Charge neutrality cannot be written as a per-step linear expectation over this vocabulary. The screen that defines it quantifies existentially over oxidation-state assignments; most elements in the split are multivalent; and collapsing to a single charge per element destroys the predicate rather than approximating it. This is a property of the constraint, not a gap in the implementation, and it bounds the class the layer addresses. Valence electron density was adopted in its place because it is computable for every element present and only partly correlated with mass density ($r = 0.725$), so a joint solve is not one constraint written twice.

A second boundary is the serialization. Over 393 structures and 1,217 element positions, a cell-first serialization makes the density budget determinable at 100% of positions and a composition-first serialization at 0%.

S10 Defects found, and how

Three classes of error were caught during this work by mechanisms worth reporting, since each would have produced a plausible-looking wrong number.

A budget generalization initially read cell-parameter token payloads as physical values when they are bin indices, which inflated volumes while still producing budgets that looked reasonable; it was caught by an equivalence check against the original single-constraint path rather than by inspection. A charge-variance comparison was made against the unconstrained arm when the claim required the sequence-level optimum: a true statement answering a different question, caught in review, and the corrected figure is stronger than the original. A crossing count in an earlier draft was a literal in the document-generating script rather than a value computed from the data, and survived a verification pass that checked every number in the same table; it is now recomputed and asserted by a test.

The general lesson we would pass on is that verifying numbers is not the same as verifying

prose. A test that recomputes a table's values will not catch the sentence describing them.